

**Example 2**

Next suppose the portfolio manager expects volatility in RLK to jump from 20 to 40%. What is the expected market impact cost for the same order with the new volatility estimate?

The only way the PM can obtain estimates from the three brokers is to provide the brokers with their proprietary volatility forecast 40% and ask the broker to re-run the scenario with this volatility estimate. But it is likely that the PM will not be willing to provide any broker with their market view and proprietary expectations for any stock.

But in the case of the aggregated pre-trade of pre-trade cost model the portfolio manager can easily re-compute the cost estimate with the new volatility estimate directly. This is as follows:

$$MI_{bp} = 793 \cdot (0.10)^{0.57} \cdot (0.40)^{0.78} \cdot (0.20)^{0.52} = 45.2 \text{ bp}$$

Portfolio managers can generate these estimates incorporating their expectations without providing proprietary information to brokers.

**Example 3**

The portfolio manager is interested in the cost of transacting an order of 7.5% ADV of stock ABC using a full day VWAP strategy. But unfortunately, the manager did not request cost estimates for this size order from their brokers. The options for the PM are to request cost estimates for ABC from their brokers but then the brokers would now know that the manager is interested in stock ABC, or interpolate cost estimates for this scenario. Unfortunately, linear interpolation is not a direct process because neither the order size nor POV rate was provided by any of the brokers. But it still can be done in three steps.

The PM can easily utilize the aggregate pre-trade model to determine the expected cost under this scenario. This calculation is:

$$MI_{bp} = 793 \cdot (0.075)^{0.57} \cdot (0.30)^{0.78} \cdot (0.0698)^{0.52} = 17.7 \text{ bp}$$

**Example 4**

A PM is evaluating a worse case scenario to liquidate a 10% ADV position of RLK under extreme situations using a full day VWAP strategy. The PM is interested in the cost to liquidate the position if volatility spikes to 40% and volume on the trade day is only half of its normal volume. Here, the POV rate for the full day VWAP strategy is  $0.10/(0.5 + 0.10) = 0.167$ .